A · Colour only

E1: only the head is trained
E1u: stage 4 as well

RGB image

EfficientViT-B1
ADE20K · stem, stages 1–3 frozen

no depth input

Stage 4 · 1/32
frozen: E1
fine-tuned: E1u (lr × 0.1)

f_2 f_3 f_4

EfficientViT-Seg head
trained · reads 1/8, 1/16 and 1/32

Segmentation · 40 classes

B · Fusion at the head

sum: E2, E2h, E2u, E10, E2c, E2cd
MLCA: E4, E5, E3c, E3cd, E4c, E4cd, E4cf, E4cfd

RGB image

EfficientViT-B1
ADE20K · stem, stages 1–3 frozen

Stage 4 · 1/32
frozen: v3 and «c» family
fine-tuned: E2u, E10

f_2 f_3 f_4

Depth

DeFM ResNet-18
frozen pretrained on depth

C3 · P4 · C5

E2, E2h, E2u, E4, E5: P3 · P4 · P5

fusion at 1/8, 1/16 and 1/32

sum | MLCA

EfficientViT-Seg head
trained · reads 1/8, 1/16 and 1/32

Segmentation · 40 classes

C · Fusion also in the encoder

sum: E11, E11d, E11r
MLCA: E3e, E4e, E4ed, E3er, E4er

RGB image

EfficientViT-B1
ADE20K · stem, stages 1–3 frozen

F' sum | MLCA

P4 · 1/16

Stage 4 · 1/32
fine-tuned · lr × 0.1

f_2 f_3 f_4

Depth

DeFM ResNet-18
frozen pretrained on depth

C3 · P4 · C5

at the head, always sums

sum

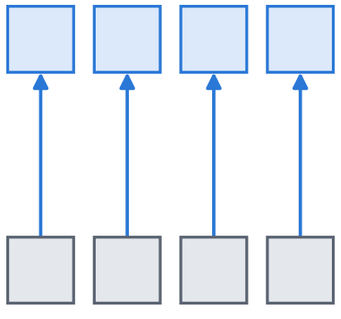
EfficientViT-Seg head
trained · reads 1/8, 1/16 and 1/32

Segmentation · 40 classes

frozen | trained | frozen or fine-tuned, depending on the experiment | sum | MLCA

Sum

colour



depth

$$g = f + W d$$

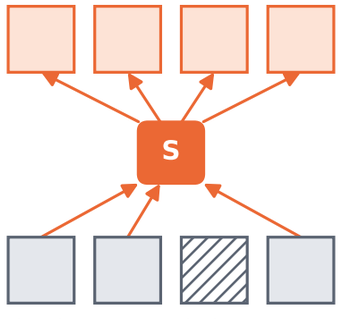
W: 1×1 convolution without bias, starts at 0 (g = f)

each position gets the depth at its own position

no mask: a hole enters as depth 0

MLCA masked linear cross-attention

colour



depth · m = 0: hole

$$g = f + W_o\, o, \qquad o_i = \varphi(q_i)^\top S \,/\, (\varphi(q_i)^\top z + \varepsilon)$$

$$S = \sum_j m_j\, \varphi(k_j)\, v_j^\top, \quad z = \sum_j m_j\, \varphi(k_j), \quad \varphi = \mathrm{ReLU}$$

q from colour; k and v from depth; W_o starts at 0

every position gets the same summary S

of the whole image, with no positional information

m = validity: binary (E4) or 1 everywhere (E3)

without depth: o = 0 and g = f